

Problem

A certain electronic test circuit produced a resonant curve with half-power points at 432 Hz and 454 Hz. If $Q = 20$, what is the resonant frequency of the circuit?

Step-by-step solution

Step 1 of 1

$$\begin{aligned} B &= \omega_2 - \omega_1 = 2\pi(f_2 - f_1) \\ &= 2\pi(454 - 432) = 44\pi \\ \omega_o &= 2\pi f_o = QB = (20)(44\pi) \\ f_o &= \frac{(20)(44\pi)}{2\pi} = 440 \text{ Hz} \end{aligned}$$